Retail Sales Analysis: Excel Data Cleaning and Power BI Dashboard

Mini Project Report | Data source: Excel generator Retail_sales_dataset.xlsx

1. Project Overview and Objective

This project covers cleaning, transforming and analysing raw retail sales data in Excel, and building an interactive two-page Power BI dashboard on top of it to produce business insights.

The objective is to demonstrate data pre-processing in Excel (raw data to clean data to pivot summaries) and then use an interactive Power BI dashboard so that sales managers can see where revenue comes from and make informed decisions about regions, products, salespeople and customer segments.

2. Data Sources

Source and timeline: A retail sales transaction dataset (Retail_sales_dataset.xlsx) with 918 order records covering orders from January 2022 to October 2026.

Domain: Retail / Sales Analytics (multi-country, four sales regions: APAC, EMEA, LATAM, NA).

The workbook has three sheets: *Raw RetailSalesData* (original data with errors), *Imputed Retail sales data* (the cleaned working copy loaded into Power BI) and *Pivot Table* (summaries by salesperson and customer type).

3. Problem Statement

- To analyse sales performance across regions, countries, cities and products to identify top-performing areas and revenue trends.
- To understand customer purchasing behaviour (New, Returning, VIP) to improve marketing and retention strategies.
- To evaluate salesperson performance so that targets and territories can be balanced.
- To identify data-quality problems in the raw sales data and show how they affect reporting.

4. Attribute (Column / Feature) Details

Attribute	Data type	Description
OrderID	String	Unique order identifier (format ORD-10001)
OrderDate	Date	Date on which the order was placed
CustomerID	String	Unique customer identifier (format CUST-1143)
ProductID	Categorical / String	Product code: P101–P103, P201–P203 (P-series) and C301–C304 (C-series)
UnitsSold	Numeric (integer)	Number of units in the order (1 to 20)
PricePerUnit	Numeric	Selling price per unit (45 to 100)
TotalRevenue	Numeric	Order value = UnitsSold × PricePerUnit
Region	Categorical	Sales region: APAC, EMEA, LATAM, NA
Country	Categorical	Country of the sale (10 countries)
City	Categorical	City of the sale (12 cities)
SalesPerson	Categorical	Salesperson who handled the order (9 named reps)
CustomerType	Categorical	New, Returning or VIP

5. Tools & Technologies

- **Excel:** data cleaning, imputation, transformation, and Pivot Tables.
- **Power BI Desktop:** data loading, data modelling, DAX measures, visualisation and interactive dashboard creation (slicers, drill-down, bookmarks).

6. Data Pre-Processing (Excel / Power Query)

6.1 Data quality issues found in the raw data

Issue	Where	Count
Missing values	Region	251 blanks/#N/A (about 27%)
Missing values	OrderDate, CustomerID, ProductID, CustomerType	30, 29, 34, 35
Missing values	UnitsSold, PricePerUnit, TotalRevenue, Country, City, SalesPerson	21, 32, 25, 25, 30, 13
Error values (#N/A)	Several columns, including Region and CustomerType	5 per affected column

Extra leading/trailing spaces	Region, Country, City, SalesPerson, ProductID, CustomerType, dates	5–12 per column, creating fake duplicate categories such as " APAC " and "APAC"
Placeholder text	OrderID and OrderDate ("UNKNOWN")	2 order IDs, 7 dates
Encoding error	City: "SÃ£o Paulo" instead of "São Paulo"	116 rows
Mixed salesperson naming	Generic "Sales Rep 1–5" mixed with named reps	5 + 4 names
Header typos in the working sheet	"Unite sold", leading spaces in " Order Id", " Date", " Product id "	4 columns

6.2 Tasks performed

- **Data cleaning and transformation:** removed or flagged invalid records (UNKNOWN order IDs), replaced missing text values with "Unknown", filled numeric gaps, standardised formats and created the imputed working sheet.
- **Filtering and sorting:** organised the data by date and by revenue to focus on relevant records.
- **Pivot Tables:** built summaries of revenue by salesperson and by customer type (Pivot Table sheet; grand total of 664,645).
- **Fact and dimension tables:** the cleaned data can be split into a fact table (OrderID, OrderDate, CustomerID, ProductID, UnitsSold, PricePerUnit, TotalRevenue) and dimension tables for Date, Product, Customer, Geography (Region, Country, City) and SalesPerson.

6.3 Full cleaning rules (raw data to clean data)

1. **Trim and standardise text** in every text column so that " APAC " and "APAC" become one value.
2. **Convert #N/A and blanks** to proper nulls, then handle them column by column.
3. **Fix encoding**: replace "SÃ£o Paulo" with "São Paulo".
4. **Fill Region from Country**: each country belongs to exactly one region (USA and Canada to NA; Brazil and Argentina to LATAM; UK, France and Germany to EMEA; India, Japan and Australia to APAC). This recovers all 251 missing regions.
5. **Fill Country and City from each other** where one is present (for example Mumbai implies India).
6. **Recompute numeric fields from the business rule** TotalRevenue = UnitsSold × PricePerUnit. In the raw data this rule holds for every complete row, so a missing value in any one of the three can be derived from the other two instead of using a mean. Using a mean produces fractional units (the working sheet shows 9,607.7 units in total).
7. **Dates:** convert text dates to the Date type and leave unrecoverable dates blank instead of inventing them.
8. **Replace** missing SalesPerson, CustomerType and ProductID with "Unknown" so these rows still count in totals.

9. **Remove duplicates:** no duplicate OrderIDs or fully duplicated rows were found.

Result after full cleaning: 918 orders, total revenue 677,843, no missing Region, and no revenue mismatches.

7. Data Modelling and DAX (Power BI)

Data model: The cleaned sheet is loaded into Power BI as one table, "Retail sales data" (12 columns). A single-table (flat) model is acceptable for this size; a star schema with the dimension tables from section 6.2 and a Date table linked one-to-many to the fact table is the recommended extension.

Calculated columns and DAX measures (the dashboard uses Sum, Average, Count and Max aggregations of these fields):

Measure	DAX
Total Revenue	Total Revenue = SUM('Retail sales data'[Total Revenue])
Total Orders	Total Orders = COUNTROWS('Retail sales data')
Units Sold	Units Sold = SUM('Retail sales data'[Unite sold])
Avg Revenue per Order	Avg Revenue per Order = DIVIDE([Total Revenue], [Total Orders])
Avg Price per Unit	Avg Price per Unit = AVERAGE('Retail sales data'[Price per unit])
Max Price per Unit	Max Price per Unit = MAX('Retail sales data'[Price per unit])
Revenue Share %	Revenue Share % = DIVIDE([Total Revenue], CALCULATE([Total Revenue], ALL('Retail sales data')))
YoY Growth %	YoY Growth % = DIVIDE([Total Revenue] - CALCULATE([Total Revenue], SAMEPERIODLASTYEAR('Date'[Date])), CALCULATE([Total Revenue], SAMEPERIODLASTYEAR('Date'[Date])))
Cumulative Revenue	Cumulative Revenue = CALCULATE([Total Revenue], FILTER(ALL('Date'[Date]), 'Date'[Date] <= MAX('Date'[Date])))

Calculated column: Year, Quarter and Month are taken from the built-in Date Hierarchy on the date column.

8. Analysis and Visualizations (Power BI)

The report has two pages with a consistent navy header (#1F2A44), light background and a clear title on every visual.

Page 1: Sales Analysis

Visual	Type	Purpose
Total Revenue, Total Orders, Units Sold, Avg Revenue per Order, Avg Price per Unit, Max Price per Unit	Cards	Headline KPIs
Revenue Trend Over Time	Line chart (drill: Year, Quarter, Month, Day)	Seasonality and growth
Revenue by Region	Donut chart	Regional share
Revenue by Salesperson	Clustered column chart	Rep performance
Revenue and Units Sold by Country	Line and clustered column combo chart	Country comparison
Revenue by Product ID	Treemap	Product contribution
Date Range, Customer Type, Region	Slicers	Interactive filtering

[SCREENSHOT OF DASHBOARD: Sales Analysis page]

Page 2: Customer & Geography Insights

Visual	Type	Purpose
Revenue by Country and City	Filled map (drill: Country to City)	Geographic spread
Revenue by City	Bar chart	City ranking
Orders by Customer Type	Funnel chart	Customer mix
Units Sold by Customer Type	Donut chart	Volume by segment
Total Revenue by Order, Product, Customer and Region	Table	Order-level detail
Revenue by Region and Customer Type	Matrix (expand Region to Order)	Cross-analysis
Date Range, Customer Type, Region	Slicers	Interactive filtering

Interactivity: slicers on both pages, drill-down on the date hierarchy, the map and the matrix, and cross-filtering between visuals. Bookmarks (for example Overview, Regional View, Customer View) and page-navigation buttons should be added from the View > Bookmarks pane.

9. Insights & Conclusions

All figures are from the fully cleaned dataset (918 orders, total revenue 677,843, 9,802 units, average order value 738).

Key findings

- **Region:** APAC leads with 188,067 (28%), followed by NA 175,639 (26%), EMEA 159,099 (23%) and LATAM 155,038 (23%).
- **Country and city:** USA is the top country (104,585), then Argentina (83,757), India (73,616), Brazil (71,281) and Canada (71,054). Buenos Aires is the top city (83,757); Germany is lowest (43,584).
- **Products:** P103 (89,740) and P101 (86,465) are the best sellers by revenue. P-series products average 838 per order against 583 for C-series.
- **Customers:** New customers generate the most revenue (230,640), then VIP (219,042) and Returning (206,945). VIP orders average 753 against 714 for Returning.
- **Salespeople:** Sales Rep 4 (97,360) and Sales Rep 5 (95,536) lead; Aarav Patel is lowest (35,780) with the fewest orders (46).
- **Trend:** Q4 is consistently the strongest quarter (for example 44,659 in 2022 Q4 and 49,019 in 2023 Q4). Annual revenue peaked in 2023 (147,345) and has been broadly flat since.

Descriptive

Revenue is spread fairly evenly across four regions, with APAC ahead. Order values range from 45 to 2,000, and a few high-value customers dominate: CUST-1076 alone contributes 11,519.

Diagnostic

- LATAM has a high order count (234, close to APAC's 244) but the lowest average order value (663), mainly because Brazil averages only 604 per order, which is driven by low-priced C-series products.
- C-series products (C301, C302) have both lower prices and lower revenue per order, which pulls down the regions where they sell most.
- About 27% of regions were blank in the raw data. Any dashboard built on uncleaned data would show regions and totals incorrectly; the working sheet still shows separate " APAC " and "APAC" values and about 3,255 of revenue under #N/A.

Predictive

The repeating Q4 peak suggests that Q4 will remain the strongest quarter, and with yearly revenue flat at roughly 127,000 to 147,000 a year, the business should expect similar results without a change in strategy. A Power BI forecast on the line chart can confirm this.

Prescriptive

- Run seasonal campaigns and stock up ahead of Q4.
- Push high-revenue P-series products (P103, P101) in LATAM and shift mix away from low-value C-series bundles to raise average order value.
- Invest in APAC and India, which have the highest average order values, and review the Germany and Japan territories.
- Set up a retention programme to convert New customers into VIP and Returning customers.
- Coach or rebalance low-volume salespeople and capture salesperson, region and customer type at entry so that blanks stop occurring.

10. Conclusions

The integration of Excel and Power BI proved effective for end-to-end data analysis, from raw data to visual reporting. Excel was used to find and fix data-quality problems and to summarise the data, and Power BI turned the clean data into an interactive dashboard that clearly shows the top regions, products, customers and sales reps, and the seasonal Q4 pattern.